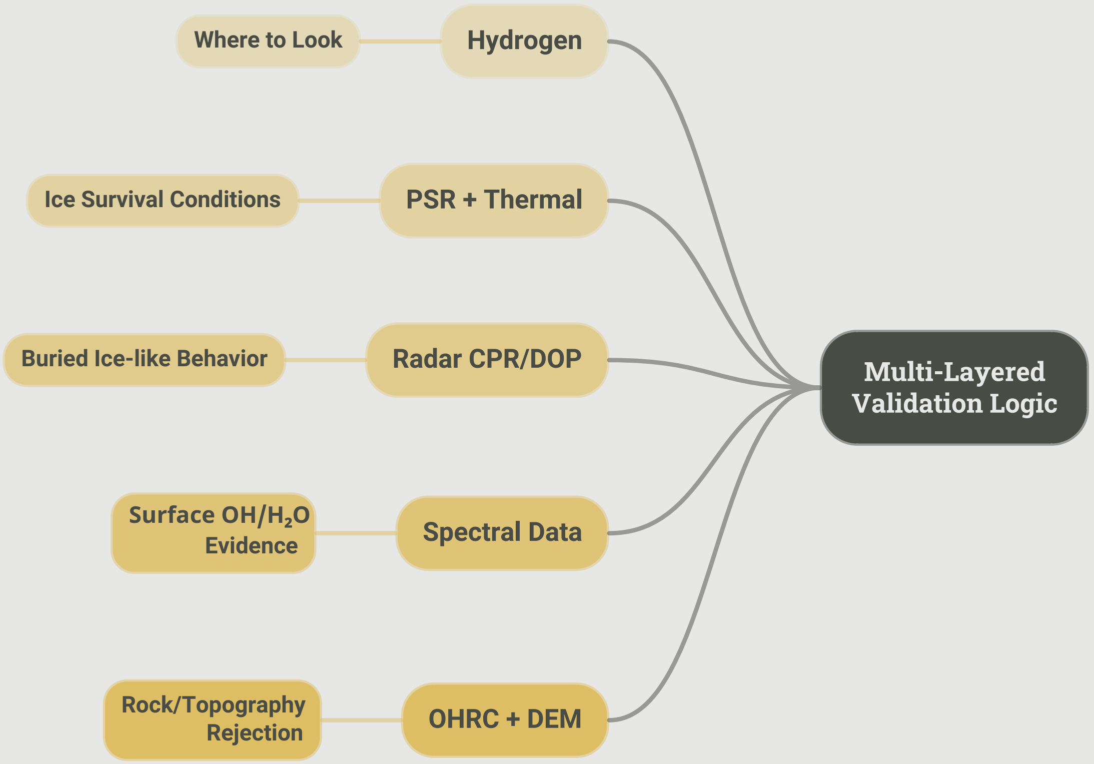
Ice Detection Strategy

From Volatile Clues to Ice Candidates: A Multi-Sensor Fusion Approach

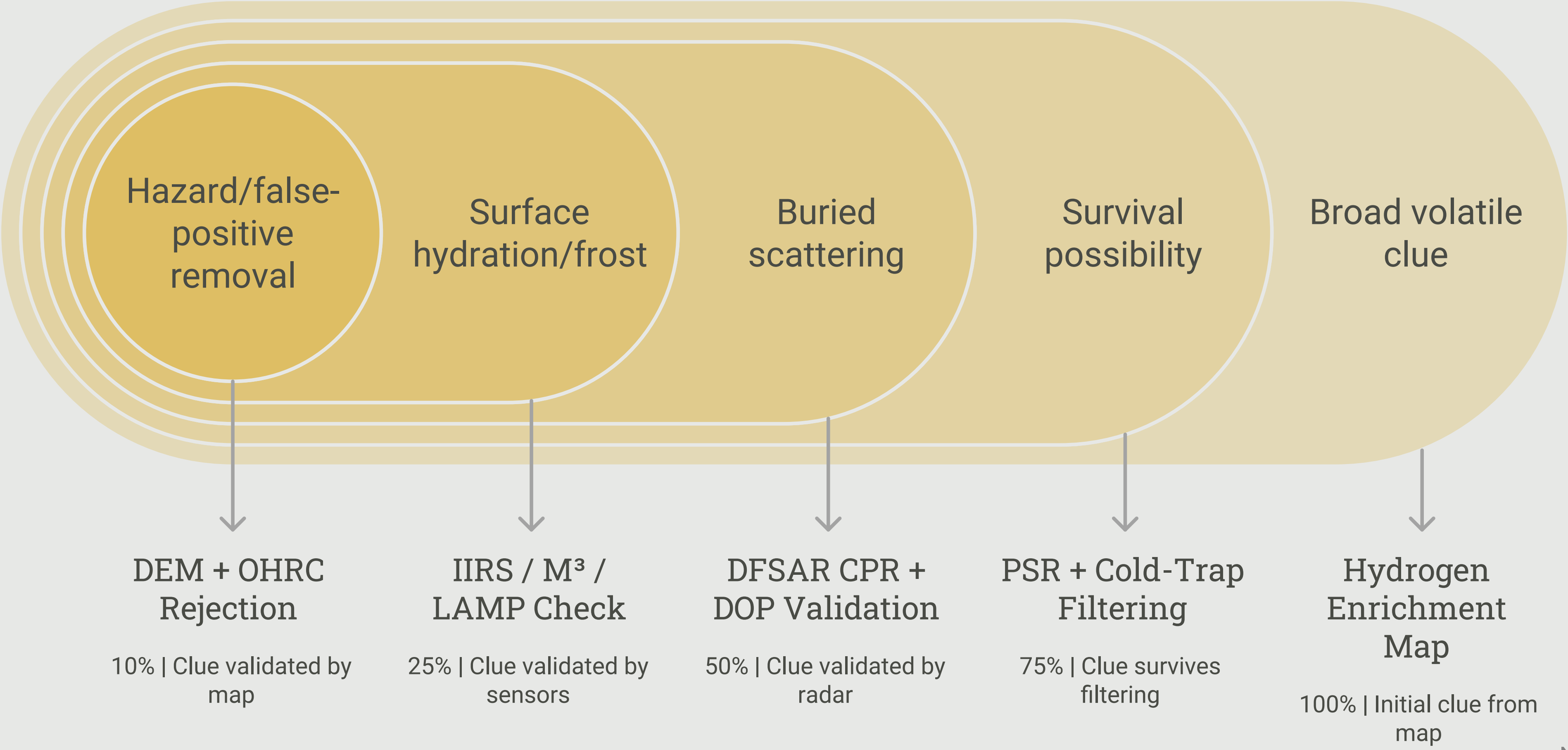
From Volatile Clues to Ice Candidates



Combining independent evidence layers to reduce false positives

We do not claim ice from a single sensor. Our strategy relies on a multi-layered validation logic to ensure accuracy, using hydrogen to identify potential sites, thermal and radar data to confirm physical conditions and burial, and optical topography to reject false positives.

Conservative Ice Confidence Pipeline



Explainable Ice Score for Mission Planning

0.20H

Ice Score Weighting

The weight assigned to the H sensor input for the final score.

0.20PSR

PSR Weighting

The weight assigned to the PSR sensor input for the final score.

0.20T

Terrain Weighting

The weight assigned to the T sensor input for the final score.

0.25R

Rover Weighting

The weight assigned to the R sensor input for the final score.

0.10Spec

Spectral Weighting

The weight assigned to the Spec sensor input for the final score.

0.05Morph

Morphological Weighting

The weight assigned to the Morph sensor input for the final score.

Selecting the Landing Site

The landing site selection process prioritizes science targets over automatic landing zones. The search focuses on identifying safe, illuminated, and accessible areas in the immediate vicinity of high-priority ice clusters.

Best Landing Site = Safe Landing + Reachable Ice

Sites are ranked by combining landing safety, operational constraints, and reachable ice evidence.

Terrain Safety		Reachable Ice
Minimizes risk of vehicle damage during descent Focuses on landing ellipse reliability and surface stability	Primary Mission Objective	**Maximizes scientific return through resource acquisition** Prioritizes high-evidence ice targets within rover range
Requires flat, boulder-free zones for touchdown Evaluates slope, roughness, and crater density	Operational Constraint	**Depends on traverse feasibility and distance weights** Calculates accessibility based on rover mobility limits

Rover Design Options for Lunar South Pole Exploration

	Solar-Optimized Rover	RTG-Based Rover
Target Region	Sunlit ridges, crater rims, PSR boundaries	Deep PSRs, cold traps, low-illumination zones
Key Benefits	Lower complexity, lighter power architecture	Sunlight-independent, continuous power
Major Challenges	Dependence on sunlight, limited PSR operation	Higher mass, cost, and regulatory complexity

Solar-optimized designs are preferred for illuminated ridges and PSR boundaries, while RTG-based systems are essential for long-duration exploration within shadowed regions.

Design-Dependent Mission Strategy

	Solar-Optimized Rover	RTG-Based Rover
Routing Priority	Safe paths with solar exposure	Safe terrain and ice targets
Power Management	Battery-aware, sunlight-based duty cycles	Continuous power budget, reduced constraints
Thermal Management	Electrical heaters, avoid long cold-soaks	Radioisotope decay heat support
Science Reach	PSR boundaries and illuminated regions	Deeper PSRs and cold traps

The selected rover design directly changes routing preference, power usage, thermal survival, and science reach.

GPU-Accelerated Lunar Path Detection Process

1. **Lunar DEM Conversion**
   Convert Lunar DEM to terrain grid
2. **Cost Value Relaxation**
   Calculate costs using Slope, Roughness, and Shadow factors
3. **GPU Cost Propagation**
   Parallelize FMM/FIM cost propagation across GPU cores
4. **Modified Theta* Extraction**
   Extract lowest-risk smooth line-of-sight path
5. **Prototype Result**
   Path generation time reduced from 15 minutes to 20 seconds

Checkpoint-Based Traverse Verification

Verification Inputs

- Rover camera feed and object detection
- Sun angle estimation
- Nearby crater and boulder patterns
- DEM-matched terrain features